

STA201 Assignment 2 (Spring 2023)

Total Marks: 30

Question 1

Blood cocaine concentration (mg/L) was determined both for a sample of individuals who had died from cocaine-induced excited delirium (ED) and for a sample of those who had died from a cocaine overdose without excited delirium; survival time for people in both groups was at most 6 hours. [9]

ED:

0	0	0	0	0.1	0.1	0.1	0.1	0.2	0.2
0.3	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.5	0.5
0.6	0.7	0.7	0.8	0.9	1	1.5	2	2.1	2.5
3	3.9	5	8.5	11	12.5	14			

NON-ED:

0	0	0	0.1	0.1	0.2	0.3	0.3	0.3	0.5
0.6	0.6	0.8	0.8	0.8	0.8	1	1.5	2	2.2
2.6	3.3	5	5.5	7	7.7	9	11	12	15
18	19.5	22	24	25.5	25.5	25.5	27	27.5	28

- Determine the three-quartile values for blood cocaine concentration for both ED and Non-ED samples. [1.5+1.5]
- Construct a comparative boxplot (two boxplots on the same set of axes, one above the other) [2]
- Describe (Compare and Contrast) the ED and Non-ED samples in your own words from the boxplot. [2]
- Are any outliers within the data set? If so, include them in the box-plot and identify them. If not, please explain why. [2]

Question 2

A Benchmark company is testing Computer Power Supply Unit (PSU) of two different companies X and Y. PSU's were tested at the 120% stress to check for the overvoltage protection. The following data given in Table, represent time to failure (in hours) for random samples taken from the days' production of the companies. [12.5]

Table : Failure time (in hours) of PSU of X and Y Company

Company	Failure time (in hours)									
X	0.54	1.80	4.52	2.05	1.03	1.18	0.80	1.33	1.29	1.11
	3.34	1.54	0.08	4.12	0.60	0.72	0.92	1.05	1.66	3.03
	1.81	2.17	0.63	0.56	5.03	0.09	0.18	0.34	1.51	1.45
Y	1.52	0.19	1.55	0.02	0.07	3.65	0.40	0.24	1.66	1.45
	1.60	1.80	4.69	0.08	7.89	1.58	0.64	0.03	0.23	0.72

- What are the sample size of the company X & Y? [0.5]
- Compute the sample mean, variance, and standard deviation for the time to failure for company X. [1+2.5+0.5]
- Compute the sample mean, variance, and standard deviation for the time to failure for company Y. [1+2.5+0.5]
- Compute the coefficient of variation (CV) for the time to failure for company X. [0.5]
- Compute the coefficient of variation (CV) for the time to failure for company Y. [0.5]
- Which measure one should consider to compare the performance / consistency of the product of the two company? And why? [2]
- Which company's product is more consistent in their performance? [1]

Question 3

You are given the following data pertaining to kilowatt hours of electricity consumed by some users in the city of Dhaka: [5.5]

Consumption (in Kilowatt hours)	Number of users
20 – 30	13
30 – 40	35
40 – 50	25
50 – 60	23
60 – 70	0 + X

Where **X** is the sum of the **last two digits** of your University ID. For example: if your university ID is: 181234**56** then **X** would be **5 + 6 = 11**. Estimate the following:

- a) sample variance and the sample standard deviation [2.5]
- b) mean deviation [2.0]
- c) coefficient of variation. [1.0]

Question 4

For a distribution Karl Pearson's coefficient of skewness is 0.64, Variance is 40 and mean is

$\sqrt{(X + Y + Z)}$ Find mode and median. [Here X,Y,Z = First 3 digits of you Student ID] [3]